

实验一

用触发器、组合函数器件和门电路设计一个广告流水灯，该流水灯由 8 个 LED 组成，工作时始终为 1 暗 7 亮，且这一个暗灯循环右移。

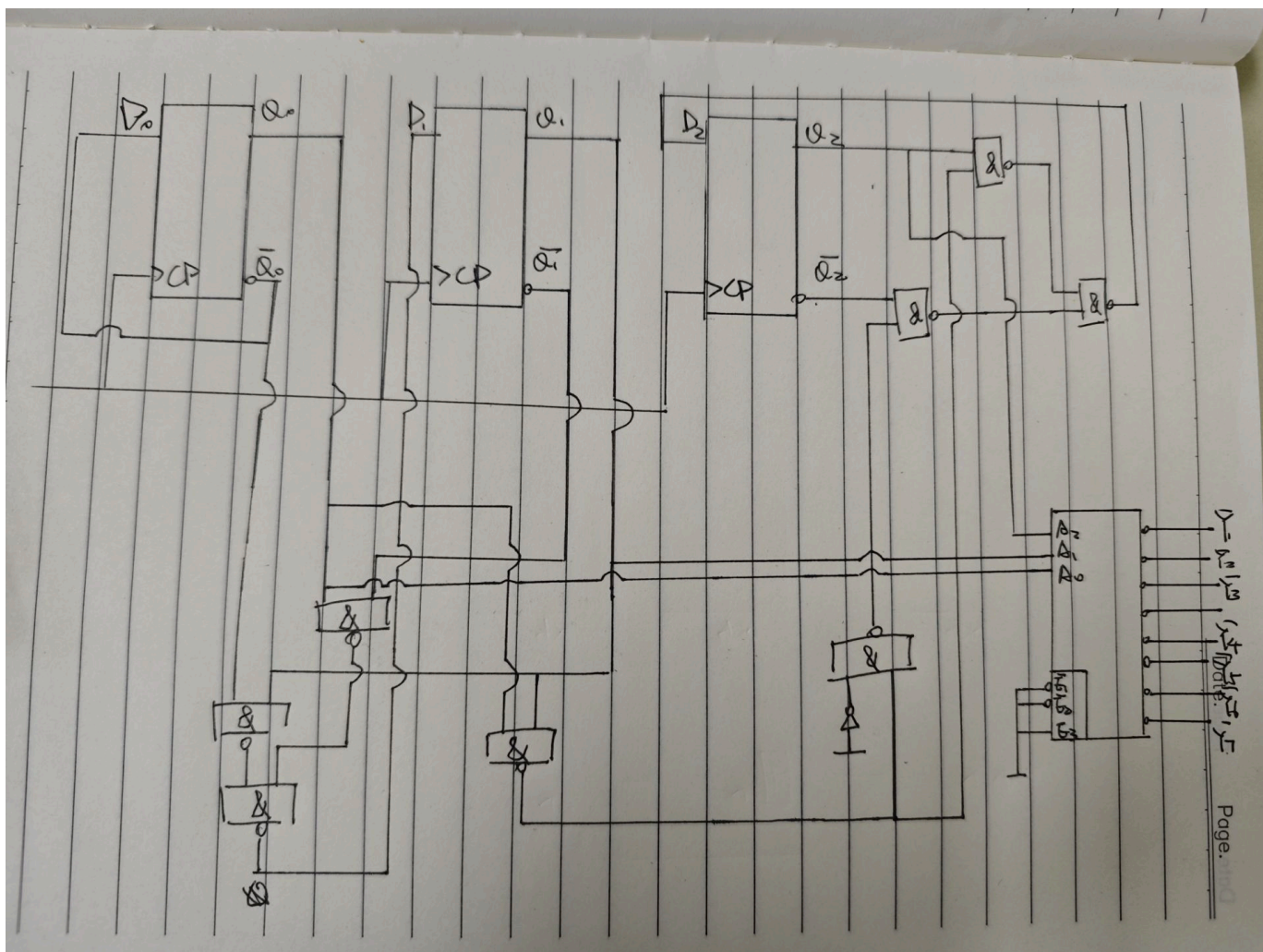
状态转移真值表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0

逻辑函数

$$\begin{aligned} Q_0^{n+1} &= \overline{Q_0^n} \\ Q_1^{n+1} &= \overline{Q_0^n} Q_1^n + \overline{Q_1^n} Q_0^n = \overline{\overline{Q_0^n} Q_1^n \cdot \overline{Q_1^n} Q_0^n} \\ Q_2^{n+1} &= \overline{Q_2^n} Q_1^n Q_0^n + Q_2^n \overline{Q_1^n} \overline{Q_0^n} = \overline{\overline{Q_2^n} Q_1^n Q_0^n \cdot Q_2^n \overline{Q_1^n} \overline{Q_0^n}} \end{aligned}$$

原理图



实物连接图

